

Question_3:

Problem Description:

You are given an array A of Q integers and Q queries. In each query, you are given an integer i ($1 \leq i \leq N$).

Your task is to find the minimum index greater than i ($1 \leq i \leq N$) such that:

1. Sum of digits of A_i is greater than the sum of digits of A_j
2. $A_i < A_j$

If there is no answer, then print -1.

Constraints

$1 \leq N, Q \leq 10^5$

$1 \leq A_i \leq 10^9$

$1 \leq Q_i \leq N$

Input format

- The first line contains two numbers N and Q .
- The next line contains N numbers.
- Next Q lines contain Q queries.

Output format

Print the answer as described in the problem

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 5
62 70 28 62 92
1
5
3
4
2
```

EXPECTED OUTPUT

```
2 -1 4 -1 -1
```

Test Case 2

INPUT (STDIN)

```
6 6
12 71 21 16 91
11
15
31
14
21
```

EXPECTED OUTPUT

```
-1 -1 -1 -1 -1
```

Code:

```
1  /*
2  Question_3:
3  For each query i, find the minimum j > i such that
4  arr[i] < arr[j] and digit sum of arr[i] is greater
5  than the digit sum of arr[j]. Print -1 if no such index exists.
6  */
7
8  #include <iostream>
9  #include <vector>
10 using namespace std;
11
12 int sumOfDigits(long long n)
13 {
14     int sum = 0;
15     while(n > 0)
16     {
17         sum += n % 10;
18         n /= 10;
19     }
20     return sum;
21 }
22 int main()
23 {
24     int N, Q;
25     cin >> N >> Q;
26
27     vector<long long> arr(N + 1);
28     vector<int> arr2(N + 1);
29
30     for(int i = 1; i <= N; i++)
31     {
32         cin >> arr[i];
33         arr2[i] = sumOfDigits(arr[i]);
34     }
35     vector<int> ans(N + 1, -1);
36     for(int i = 1; i <= N; i++)
37     {
38         for(int j = i + 1; j <= N; j++)
```

```
39         {
40             int x = i;
41             int y = j;
42             if(arr[x]<arr[y])
43             {
44                 if(arr2[x]>arr2[y])
45                 {
46                     ans[i] = j;
47                     break;
48                 }
49             }
50         }
51     }
52     for(int q = 0; q < Q; q++)
53     {
54         int idx;
55         cin >> idx;
56
57         if(idx >= 1 && idx <= N)
58             cout << ans[idx];
59         else
60             cout << -1;
61         if(q < Q - 1)
62             cout << " ";
63     }
64     cout << endl;
65     return 0;
66 }
```

Output:

Test Case_1:

```
5 5
62 70 28 62 92
1
5
3
4
2
2 -1 4 -1 -1
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
6 6
12 71 21 16 91 41
11
15
31
14
21
34
-1 -1 -1 -1 -1 -1
PS C:\Users\Prabin Yadav\
```